Cache Simulator - Test Report

1. Test Objective

Validate functional correctness of cache operations against project requirements, including hit/miss behavior, MESI transitions, LRU/invalid handling, snoop/invalidate behavior, reset behavior, and instruction/data cache separation.

2. Test Environment

Executable: app\\cache_sim.exe

Input traces: inputs\\tests*.txt

Execution mode for verification: MODE=1 (includes L2 communication logs)

3. Test Execution Method

Each trace file was executed with: app\\cache_sim.exe <trace_file> 1

4. Test Catalog

1. t01_data_read_miss_hit.txt
Checks data-read hit/miss flow for opcode 0.

- First read must be a miss and fetch from L2.
- Second read to same line must be a hit.
- MESI should move from E to S on the read hit (project-specific assumption).

2. t02_data_write_hit_mesi.txt
Checks write-hit behavior for opcode 1 after a read allocation.

- Read allocates line in E.
- Write hit transitions line to M and marks dirty.
- Repeated write should stay M (no new allocation).

3. t03_write_miss_allocate.txt
Checks write-miss allocation path.

- Write miss should do Read-For-Ownership behavior and install line in M.
- Dump should show one valid dirty line for that tag.

4. t04_invalidate_dirty_line.txt
Checks opcode 3 invalidate semantics.

- Start with dirty M line.
- Invalidate should write back to L2 (mode 1 log), then set line invalid.
- Final dump should not list that line.

5. t05_snoop_dirty_line.txt
Checks opcode 4 snoop semantics.

- Start with dirty M line.
- Snoop should return data to L2 (mode 1 log), then invalidate local line.

6. t06_invalid_preferred_over_lru.txt
Checks prefer invalid line before eviction requirement.

- Fill all 8 ways of one set.
- Invalidate one way.
- Next write miss must use invalid slot, not evict another valid line.

7. t07_dirty_eviction_writeback.txt
Checks eviction and writeback behavior when set is full.

- Fill set with 8 dirty lines (write misses).
- 9th write miss forces eviction.
- Evicted dirty victim should be written back to L2 (mode 1 log).

8. t08_instruction_cache_basic.txt
Checks opcode 2 instruction-read path.

- First instruction read miss allocates in L1I.
- Second instruction read hit should be visible in L1I stats/dump.
- Confirms L1I and L1D are tracked separately.

9. t09_reset_clears_all.txt
Checks opcode 8 reset behavior.

- Perform both data and instruction accesses.
- Dump before reset should show entries.
- Dump after reset should show empty caches and reset stats.

10. t10_full_opcode_smoke.txt
Quick smoke test across major opcodes.

- Exercises 0,1,2,3,4,8,9 in one trace.
- Useful for a fast regression check after code changes.

11. scenario_s1_fill_set_stepdump.txt
Goal:

- Fill one L1D set (index 0x3784) using 8 read misses.
- Verify way fill order, MESI allocation state, and initial LRU progression.

Expected per step:

1. Each 0 <addr> is a miss (Read from L2 in mode 1).
2. Inserted line MESI = E.
3. Final set has tags in way order:
    - way0 0x984
    - way1 0x116
    - way2 0x100
    - way3 0x999
    - way4 0x645
    - way5 0x846

    - way6 0x211
    - way7 0x777
4. Final LRU should be ordered oldest->newest as 000..111 across way0..way7.

12. scenario_s2_rw_snoop_stepdump.txt

Trace: inputs/tests/scenario_pdf_s2_rw_snoop_stepdump.txt

Goal:

- Exercise key MESI transitions and replacement behavior from the PDF-style sequence.

Expected highlights by operation (after preload):

1. 0 999DE132 hit: 999 transitions E -> S.
2. 1 116DE123 hit: 116 transitions to M, dirty set.
3. 1 666DE135 miss: LRU victim replaced, new 666 in M.
4. 1 333DE12C miss: next LRU victim replaced, new 333 in M.
5. 0 846DE10C hit: 846 transitions E -> S.
6. 0 777DE136 hit: 777 transitions E -> S.
7. 1 ABCDE128 miss: replace next victim, new ABC in M.
8. 0 116DE101 hit: 116 remains M on read (project assumption).
9. 1 100DE101 miss: allocate/replace and set M.
10. 1 AAADE101 miss: allocate/replace and set M.
11. 1 EDCDE101 miss: if victim dirty/M, writeback to L2 then install EDC in M.
12. 4 AAADE101 snoop: if AAA is local dirty/M, return/writeback then invalidate AAA.

Expected final tag/state set after step 12:

- EDC:M
- 116:M
- 333:M
- AAA:I (invalid, should not appear in valid-only dump)
- ABC:M
- 846:S
- 100:M
- 777:S

13. scenario_s3_invalid_preferred_stepdump.txt

Goal:

- Validate PDF note: after a line becomes invalid, next miss should use invalid line before evicting valid useful data.

Sequence detail:

- Replays Scenario S2 through 4 AAADE101, making AAA invalid.
- Adds 1 FFFDE101 (write miss) and dumps.

Expected result:

1. FFF should occupy the invalid slot previously held by AAA.
2. A different valid line should NOT be evicted for this allocation.
3. 333 (and other still-valid lines) should remain present if they were valid before the FFF miss.

5. Detailed Test Cases, Expectations, and Results

Test ID: T01
Trace File: t01_data_read_miss_hit.txt
Objective: Data read miss then hit
Expectation: First read miss (L2 fetch), second read hit, read-hit transition E->S.
Result: Pass
Observation: Dump shows tag 984 in S state; stats reads=2, hits=1, misses=1

Test ID: T02
Trace File: t02_data_write_hit_mesi.txt
Objective: Write-hit MESI behavior
Expectation: Read allocates E; write hit transitions to M and marks dirty; repeat write remains M.
Result: Pass
Observation: Dump shows tag 116 as M and dirty; expected counters observed.

Test ID: T03
Trace File: t03_write_miss_allocate.txt
Objective: Write-miss allocation behavior
Expectation: Write miss requests ownership and installs line in M/dirty.
Result: Pass
Observation: RFO and write logs present; dump shows tag 666 as M dirty.

Test ID: T04
Trace File: t04_invalidate_dirty_line.txt
Objective: Opcode 3 invalidate dirty line
Expectation: Dirty/M line writes back to L2 then becomes invalid and disappears from valid-line dump.
Result: Pass
Observation: Write-to-L2 logged for AAA block; tag AAA absent in final dump.

Test ID: T05
Trace File: t05_snoop_dirty_line.txt
Objective: Opcode 4 snoop on dirty line
Expectation: Return data to L2 for dirty/M line and invalidate local copy.
Result: Pass
Observation: Return-data log seen for EDC block; line absent from final dump.

Test ID: T06
Trace File: t06_invalid_preferred_over_lru.txt
Objective: Prefer invalid line over evicting valid line
Expectation: After creating invalid slot, new miss fills invalid slot first.
Result: Pass

Observation: AAA inserted into invalid way; previously valid lines retained.
Test ID: T07 **Trace File:** t07_dirty_eviction_writeback.txt **Objective:** Dirty eviction behavior on full set **Expectation:** 9th write miss causes eviction of LRU dirty line with writeback. **Result:** Pass **Observation:** Writeback logged for evicted 984 line; EDC installed as M.
Test ID: T08 **Trace File:** t08_instruction_cache_basic.txt **Objective:** Instruction cache path **Expectation:** Opcode 2 affects L1I independently with miss then hit. **Result:** Pass **Observation:** L1I shows tag 984 in dump; L1I stats reads=2, hits=1, misses=1.
Test ID: T09 **Trace File:** t09_reset_clears_all.txt **Objective:** Reset behavior **Expectation:** Opcode 8 clears L1I/L1D contents and resets all counters. **Result:** Pass **Observation:** Post-reset dump empty; final stats for both caches are zero.
Test ID: T10 **Trace File:** t10_full_opcode_smoke.txt **Objective:** Smoke test across core opcodes **Expectation:** Exercise 0/1/2/3/4/8/9 and observe corresponding logs/dumps. **Result:** Pass **Observation:** All target operations produced expected observable effects in mode 1 output.
Test ID: S1 **Trace File:** scenario_s1_fill_set_stepdump.txt **Objective:** Fill one L1D set using 8 read misses **Expectation:** All 8 reads miss; MESI allocates as E; final tags in expected way order; LRU progresses oldest → newest (000..111). **Result:** Pass **Observation:** Observed 8 Read-from-L2 logs; final dump shows tags 984, 116, 100, 999, 645, 846, 211, 777 with E states and LRU 000..111.
Test ID: S2 **Trace File:** scenario_s2_rw_snoop_stepdump.txt **Objective:** Validate PDF-style read/write/replace/snoop sequence **Expectation:** E → S on read hit, writes move lines to M, misses trigger replacement, opcode4 returns data for dirty line and invalidates it. **Result:** Pass **Observation:** All expected transitions observed; final set contains EDC:M, 116:M, 333:M, ABC:M , 846:S, 100:M, 777:S; AAA invalid after snoop.

Test ID: S3 **Trace File:** scenario_s3_invalid_preferred_stepdump.txt **Objective:** Validate invalid-line preference before LRU eviction **Expectation:** After AAA invalidation, next miss should allocate into invalid slot, not evict another valid line. **Result:** Pass **Observation:** Write miss for FFF allocated in way previously occupied by invalid AAA; other valid lines remained present.

6. Summary

Total tests executed: 13

Passed: 13

Failed: 0

Conclusion: Current implementation satisfies the tested project requirements for scenario-based behavior and unit-style trace validation.

7. Test Execution

```
============================================================
Trace File : t01_data_read_miss_hit.txt
Mode       : 1
============================================================
  [L2] Read from L2           0x984DE132

==== Cache Dump @ Trace Line 5 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   S    111  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 2
  Writes  : 0
  Hits    : 1
  Misses  : 1
  HitRate : 50.00%
============================================================

============================================================
Trace File : t02_data_write_hit_mesi.txt
Mode       : 1
============================================================
  [L2] Read from L2           0x116DE12F
  [L2] Write to L2            0x116DE123

==== Cache Dump @ Trace Line 6 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
```

```
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x116   M    111  1
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 1
  Writes  : 2
  Hits    : 2
  Misses  : 1
  HitRate : 66.67%
============================================================

============================================================
Trace File : t03_write_miss_allocate.txt
Mode       : 1
============================================================
  [L2] Read for Ownership from L2 0x666DE135
  [L2] Write to L2            0x666DE135

==== Cache Dump @ Trace Line 4 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x666   M    111  1
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 0
  Writes  : 1
  Hits    : 0
  Misses  : 1
  HitRate : 0.00%
============================================================
```

```
============================================================
Trace File : t04_invalidate_dirty_line.txt
Mode       : 1
============================================================
  [L2] Read for Ownership from L2 0xAAADE101
  [L2] Write to L2            0xAAADE101
  [L2] Write to L2            0xAAADE100

==== Cache Dump @ Trace Line 5 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 0
  Writes  : 1
  Hits    : 0
  Misses  : 1
  HitRate : 0.00%
============================================================

============================================================
Trace File : t05_snoop_dirty_line.txt
Mode       : 1
============================================================
  [L2] Read for Ownership from L2 0xEDCDE101
  [L2] Write to L2            0xEDCDE101
  [L2] Return data to L2      0xEDCDE100

==== Cache Dump @ Trace Line 5 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
```

```
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 0
  Writes  : 1
  Hits    : 0
  Misses  : 1
  HitRate : 0.00%
============================================================

============================================================
Trace File : t06_invalid_preferred_over_lru.txt
Mode       : 1
============================================================
  [L2] Read from L2           0x984DE132
  [L2] Read from L2           0x116DE12F
  [L2] Read from L2           0x100DE130
  [L2] Read from L2           0x999DE12E
  [L2] Read from L2           0x645DE10A
  [L2] Read from L2           0x846DE107
  [L2] Read from L2           0x211DE128
  [L2] Read from L2           0x777DE133
  [L2] Read for Ownership from L2 0xAAADE101
  [L2] Write to L2            0xAAADE101

==== Cache Dump @ Trace Line 13 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   E    000  0
  0x3784   1    0x116   E    000  0
  0x3784   2    0x100   E    001  0
  0x3784   3    0xAAA   M    111  1
  0x3784   4    0x645   E    011  0
  0x3784   5    0x846   E    100  0
  0x3784   6    0x211   E    101  0
  0x3784   7    0x777   E    110  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
```

```
L1I stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 8
  Writes  : 1
  Hits    : 0
  Misses  : 9
  HitRate : 0.00%
============================================================

============================================================
Trace File : t07_dirty_eviction_writeback.txt
Mode       : 1
============================================================
  [L2] Read for Ownership from L2 0x984DE132
  [L2] Write to L2            0x984DE132
  [L2] Read for Ownership from L2 0x116DE123
  [L2] Write to L2            0x116DE123
  [L2] Read for Ownership from L2 0x100DE101
  [L2] Write to L2            0x100DE101
  [L2] Read for Ownership from L2 0x999DE12E
  [L2] Write to L2            0x999DE12E
  [L2] Read for Ownership from L2 0x645DE10A
  [L2] Write to L2            0x645DE10A
  [L2] Read for Ownership from L2 0x846DE107
  [L2] Write to L2            0x846DE107
  [L2] Read for Ownership from L2 0x211DE128
  [L2] Write to L2            0x211DE128
  [L2] Read for Ownership from L2 0x777DE133
  [L2] Write to L2            0x777DE133
  [L2] Read for Ownership from L2 0xEDCDE101
  [L2] Write to L2            0x984DE100
  [L2] Write to L2            0xEDCDE101

==== Cache Dump @ Trace Line 12 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0xEDC   M    111  1
  0x3784   1    0x116   M    000  1
  0x3784   2    0x100   M    001  1
  0x3784   3    0x999   M    010  1
  0x3784   4    0x645   M    011  1
  0x3784   5    0x846   M    100  1
  0x3784   6    0x211   M    101  1
```

```
  0x3784   7    0x777   M    110  1
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 0
  Writes  : 9
  Hits    : 0
  Misses  : 9
  HitRate : 0.00%
============================================================

============================================================
Trace File : t08_instruction_cache_basic.txt
Mode       : 1
============================================================
  [L2] Read from L2           0x984DE132

==== Cache Dump @ Trace Line 5 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   S    111  0
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
  Reads   : 2
  Writes  : 0
  Hits    : 1
  Misses  : 1
  HitRate : 50.00%
L1D stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
============================================================

============================================================
```

```
Trace File : t09_reset_clears_all.txt
Mode       : 1
============================================================
  [L2] Read from L2           0x984DE132
  [L2] Read from L2           0x116DE12F

==== Cache Dump @ Trace Line 5 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x116   E    111  0
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   E    111  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Cache Dump @ Trace Line 7 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
============================================================

============================================================
Trace File : t10_full_opcode_smoke.txt
Mode       : 1
============================================================
  [L2] Read from L2           0x999DE132
  [L2] Read for Ownership from L2 0x116DE123
```

```
  [L2] Write to L2            0x116DE123
  [L2] Read from L2           0x846DE10C
  [L2] Return data to L2      0x116DE100

==== Cache Dump @ Trace Line 8 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x846   E    111  0
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Cache Dump @ Trace Line 10 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
============================================================

============================================================
Trace File : scenario_s1_fill_set_stepdump.txt
Mode       : 1
============================================================
  [L2] Read from L2           0x984DE132

==== Cache Dump @ Trace Line 5 ====

L1I contents (valid lines only):
```

```
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   E    111  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read from L2           0x116DE12F

==== Cache Dump @ Trace Line 7 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   E    110  0
  0x3784   1    0x116   E    111  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read from L2           0x100DE130

==== Cache Dump @ Trace Line 9 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   E    101  0
  0x3784   1    0x116   E    110  0
  0x3784   2    0x100   E    111  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read from L2           0x999DE12E

==== Cache Dump @ Trace Line 11 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
```

```
  0x3784   0    0x984   E    100  0
  0x3784   1    0x116   E    101  0
  0x3784   2    0x100   E    110  0
  0x3784   3    0x999   E    111  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read from L2           0x645DE10A

==== Cache Dump @ Trace Line 13 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   E    011  0
  0x3784   1    0x116   E    100  0
  0x3784   2    0x100   E    101  0
  0x3784   3    0x999   E    110  0
  0x3784   4    0x645   E    111  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read from L2           0x846DE107

==== Cache Dump @ Trace Line 15 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   E    010  0
  0x3784   1    0x116   E    011  0
  0x3784   2    0x100   E    100  0
  0x3784   3    0x999   E    101  0
  0x3784   4    0x645   E    110  0
  0x3784   5    0x846   E    111  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read from L2           0x211DE128

==== Cache Dump @ Trace Line 17 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
```

```
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   E    001  0
  0x3784   1    0x116   E    010  0
  0x3784   2    0x100   E    011  0
  0x3784   3    0x999   E    100  0
  0x3784   4    0x645   E    101  0
  0x3784   5    0x846   E    110  0
  0x3784   6    0x211   E    111  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read from L2           0x777DE133

==== Cache Dump @ Trace Line 19 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   E    000  0
  0x3784   1    0x116   E    001  0
  0x3784   2    0x100   E    010  0
  0x3784   3    0x999   E    011  0
  0x3784   4    0x645   E    100  0
  0x3784   5    0x846   E    101  0
  0x3784   6    0x211   E    110  0
  0x3784   7    0x777   E    111  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 8
  Writes  : 0
  Hits    : 0
  Misses  : 8
  HitRate : 0.00%
============================================================

============================================================
Trace File : scenario_s2_rw_snoop_stepdump.txt
Mode       : 1
============================================================
  [L2] Read from L2           0x984DE132
  [L2] Read from L2           0x116DE12F
```

```
  [L2] Read from L2           0x100DE130
  [L2] Read from L2           0x999DE12E
  [L2] Read from L2           0x645DE10A
  [L2] Read from L2           0x846DE107
  [L2] Read from L2           0x211DE128
  [L2] Read from L2           0x777DE133

==== Cache Dump @ Trace Line 15 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   E    000  0
  0x3784   1    0x116   E    001  0
  0x3784   2    0x100   E    010  0
  0x3784   3    0x999   S    111  0
  0x3784   4    0x645   E    011  0
  0x3784   5    0x846   E    100  0
  0x3784   6    0x211   E    101  0
  0x3784   7    0x777   E    110  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Write to L2            0x116DE123

==== Cache Dump @ Trace Line 17 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x984   E    000  0
  0x3784   1    0x116   M    111  1
  0x3784   2    0x100   E    001  0
  0x3784   3    0x999   S    110  0
  0x3784   4    0x645   E    010  0
  0x3784   5    0x846   E    011  0
  0x3784   6    0x211   E    100  0
  0x3784   7    0x777   E    101  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read for Ownership from L2 0x666DE135
  [L2] Write to L2            0x666DE135

==== Cache Dump @ Trace Line 19 ====

L1I contents (valid lines only):
```

```
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x666   M    111  1
  0x3784   1    0x116   M    110  1
  0x3784   2    0x100   E    000  0
  0x3784   3    0x999   S    101  0
  0x3784   4    0x645   E    001  0
  0x3784   5    0x846   E    010  0
  0x3784   6    0x211   E    011  0
  0x3784   7    0x777   E    100  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read for Ownership from L2 0x333DE12C
  [L2] Write to L2            0x333DE12C

==== Cache Dump @ Trace Line 21 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x666   M    110  1
  0x3784   1    0x116   M    101  1
  0x3784   2    0x333   M    111  1
  0x3784   3    0x999   S    100  0
  0x3784   4    0x645   E    000  0
  0x3784   5    0x846   E    001  0
  0x3784   6    0x211   E    010  0
  0x3784   7    0x777   E    011  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Cache Dump @ Trace Line 23 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x666   M    101  1
  0x3784   1    0x116   M    100  1
  0x3784   2    0x333   M    110  1
  0x3784   3    0x999   S    011  0
```

```
  0x3784   4    0x645   E    000  0
  0x3784   5    0x846   S    111  0
  0x3784   6    0x211   E    001  0
  0x3784   7    0x777   E    010  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Cache Dump @ Trace Line 25 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x666   M    100  1
  0x3784   1    0x116   M    011  1
  0x3784   2    0x333   M    101  1
  0x3784   3    0x999   S    010  0
  0x3784   4    0x645   E    000  0
  0x3784   5    0x846   S    110  0
  0x3784   6    0x211   E    001  0
  0x3784   7    0x777   S    111  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read for Ownership from L2 0xABCDE128
  [L2] Write to L2            0xABCDE128

==== Cache Dump @ Trace Line 27 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x666   M    011  1
  0x3784   1    0x116   M    010  1
  0x3784   2    0x333   M    100  1
  0x3784   3    0x999   S    001  0
  0x3784   4    0xABC   M    111  1
  0x3784   5    0x846   S    101  0
  0x3784   6    0x211   E    000  0
  0x3784   7    0x777   S    110  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Cache Dump @ Trace Line 29 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
```

```
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x666   M    010  1
  0x3784   1    0x116   M    111  1
  0x3784   2    0x333   M    011  1
  0x3784   3    0x999   S    001  0
  0x3784   4    0xABC   M    110  1
  0x3784   5    0x846   S    100  0
  0x3784   6    0x211   E    000  0
  0x3784   7    0x777   S    101  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read for Ownership from L2 0x100DE101
  [L2] Write to L2            0x100DE101

==== Cache Dump @ Trace Line 31 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x666   M    001  1
  0x3784   1    0x116   M    110  1
  0x3784   2    0x333   M    010  1
  0x3784   3    0x999   S    000  0
  0x3784   4    0xABC   M    101  1
  0x3784   5    0x846   S    011  0
  0x3784   6    0x100   M    111  1
  0x3784   7    0x777   S    100  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read for Ownership from L2 0xAAADE101
  [L2] Write to L2            0xAAADE101

==== Cache Dump @ Trace Line 33 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0x666   M    000  1
  0x3784   1    0x116   M    101  1
  0x3784   2    0x333   M    001  1
```

```
  0x3784   3    0xAAA   M    111  1
  0x3784   4    0xABC   M    100  1
  0x3784   5    0x846   S    010  0
  0x3784   6    0x100   M    110  1
  0x3784   7    0x777   S    011  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Read for Ownership from L2 0xEDCDE101
  [L2] Write to L2            0x666DE100
  [L2] Write to L2            0xEDCDE101

==== Cache Dump @ Trace Line 35 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0xEDC   M    111  1
  0x3784   1    0x116   M    100  1
  0x3784   2    0x333   M    000  1
  0x3784   3    0xAAA   M    110  1
  0x3784   4    0xABC   M    011  1
  0x3784   5    0x846   S    001  0
  0x3784   6    0x100   M    101  1
  0x3784   7    0x777   S    010  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====
  [L2] Return data to L2      0xAAADE100

==== Cache Dump @ Trace Line 37 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0xEDC   M    111  1
  0x3784   1    0x116   M    100  1
  0x3784   2    0x333   M    000  1
  0x3784   4    0xABC   M    011  1
  0x3784   5    0x846   S    001  0
  0x3784   6    0x100   M    101  1
  0x3784   7    0x777   S    010  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
```

```
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 12
  Writes  : 7
  Hits    : 5
  Misses  : 14
  HitRate : 26.32%
============================================================

============================================================
Trace File : scenario_s3_invalid_preferred_stepdump.txt
Mode       : 1
============================================================
  [L2] Read from L2           0x984DE132
  [L2] Read from L2           0x116DE12F
  [L2] Read from L2           0x100DE130
  [L2] Read from L2           0x999DE12E
  [L2] Read from L2           0x645DE10A
  [L2] Read from L2           0x846DE107
  [L2] Read from L2           0x211DE128
  [L2] Read from L2           0x777DE133
  [L2] Write to L2            0x116DE123
  [L2] Read for Ownership from L2 0x666DE135
  [L2] Write to L2            0x666DE135
  [L2] Read for Ownership from L2 0x333DE12C
  [L2] Write to L2            0x333DE12C
  [L2] Read for Ownership from L2 0xABCDE128
  [L2] Write to L2            0xABCDE128
  [L2] Read for Ownership from L2 0x100DE101
  [L2] Write to L2            0x100DE101
  [L2] Read for Ownership from L2 0xAAADE101
  [L2] Write to L2            0xAAADE101
  [L2] Read for Ownership from L2 0xEDCDE101
  [L2] Write to L2            0x666DE100
  [L2] Write to L2            0xEDCDE101
  [L2] Return data to L2      0xAAADE100
  [L2] Read for Ownership from L2 0xFFFDE101
  [L2] Write to L2            0xFFFDE101

==== Cache Dump @ Trace Line 26 ====

L1I contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  ----     ---- ----   ---- ---- -----

L1D contents (valid lines only):
  Set      Way  Tag    MESI LRU  Dirty
  ----     ---- ----   ---- ---- -----
  0x3784   0    0xEDC   M    110  1
  0x3784   1    0x116   M    011  1
```

```
  0x3784   2    0x333   M    000  1
  0x3784   3    0xFFF   M    111  1
  0x3784   4    0xABC   M    010  1
  0x3784   5    0x846   S    000  0
  0x3784   6    0x100   M    100  1
  0x3784   7    0x777   S    001  0
  ----     ---- ----   ---- ---- -----
==== End Dump ====

==== Final Summary ====
L1I stats:
  Reads   : 0
  Writes  : 0
  Hits    : 0
  Misses  : 0
  HitRate : 0.00%
L1D stats:
  Reads   : 12
  Writes  : 8
  Hits    : 5
  Misses  : 15
  HitRate : 25.00%
============================================================
```